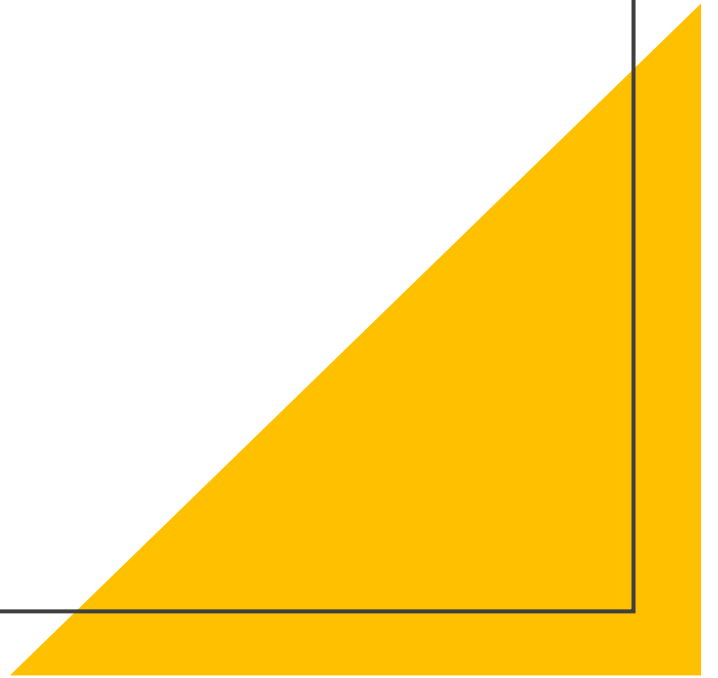


User Interface Design

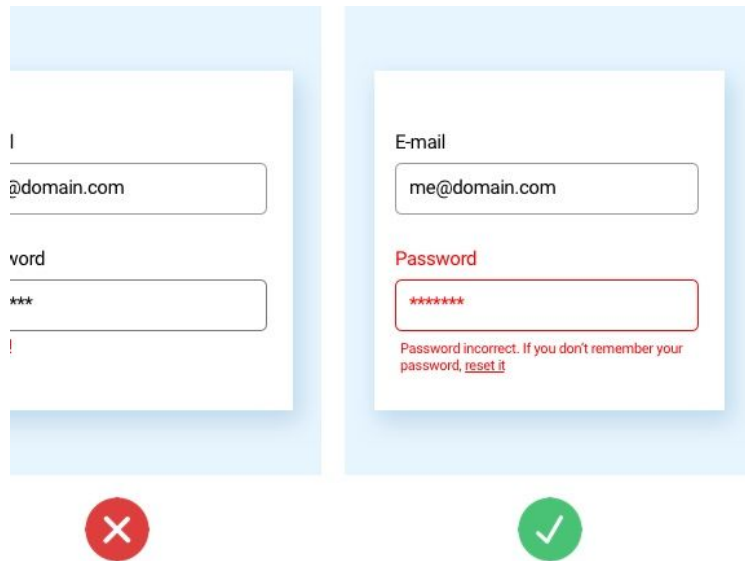




Interface Design

- Initially users must adapt to complex systems
 - For being technical
- Now technology must consider human ease
- Usability matters a lot!!!!

Interface Design



The image shows two login form designs side-by-side. The left design is labeled with a red 'X' icon, indicating it is a bad example. It has a single input field for the email address, with the text '@domain.com' inside. Below it is a password field with the text 'vord' and '***' inside. The right design is labeled with a green checkmark icon, indicating it is a good example. It has two input fields: one for the email address with the text 'me@domain.com', and one for the password with the text '*****'. Below the password field, there is a message: 'Password incorrect. If you don't remember your password, [reset it](#)'.

- Set of golden rules defined
- Interaction mechanism created
 - **the window world**
- Issues observed in bad GUI designs:
 - Not user friendly
 - Confusing
 - Counterintuitive
 - frustrating

Golden rules for UI design (by Theo Mandel)

1

Place the user in control

2

Reduce user's memory load

3

Make a consistent interface

Principle 1: Allow users to maintain control

- **Define interactive modes that allow the user to let them do whatever they want freely**
 - Define different modes and allows easy switching between modes
 - E.g., spell check and editing modes in the word processor, read mode, browser modes, etc.
- **Allow flexible interaction**
 - Allow users to interact via multiple ways like the mouse, keyboard, digitizer pen, voice inputs, multitouch inputs, etc.
- **Allow user interaction to be uninterruptable and undoable**
 - Freely allows the user to quit a work and do another without losing changes
 - Undo/redo also allowed

Principle 1: Allow users to maintain control

- **Allow creating customized operations**
 - Create macros for Frequently occurring operations(e.g., like in MS word)
 - For advanced users to customize UI
- **Hide technical internals from casual users**
 - User is unaware of the underlying OS, file management operations and other complexities
 - E.g., in Application software, there is no need to give OS commands.
- **Design for direct interaction with on-screen objects**
 - Icons or other objects could be dragged or scaled up or down as if they exist physically.

Principle 2: Reduce User's memory load

- **Well designed system doesn't test user's memory**
 - E.g., Use list boxes or checkbox inputs rather than text boxes.
- **Reduce demand to learn more**
 - Users should not have to recall past inputs, rather they are saved and shown using visuals.
- **Establish meaningful defaults**
 - User should have a preference to customize layout or change the default settings.
 - However, reset could change them to defaults again.

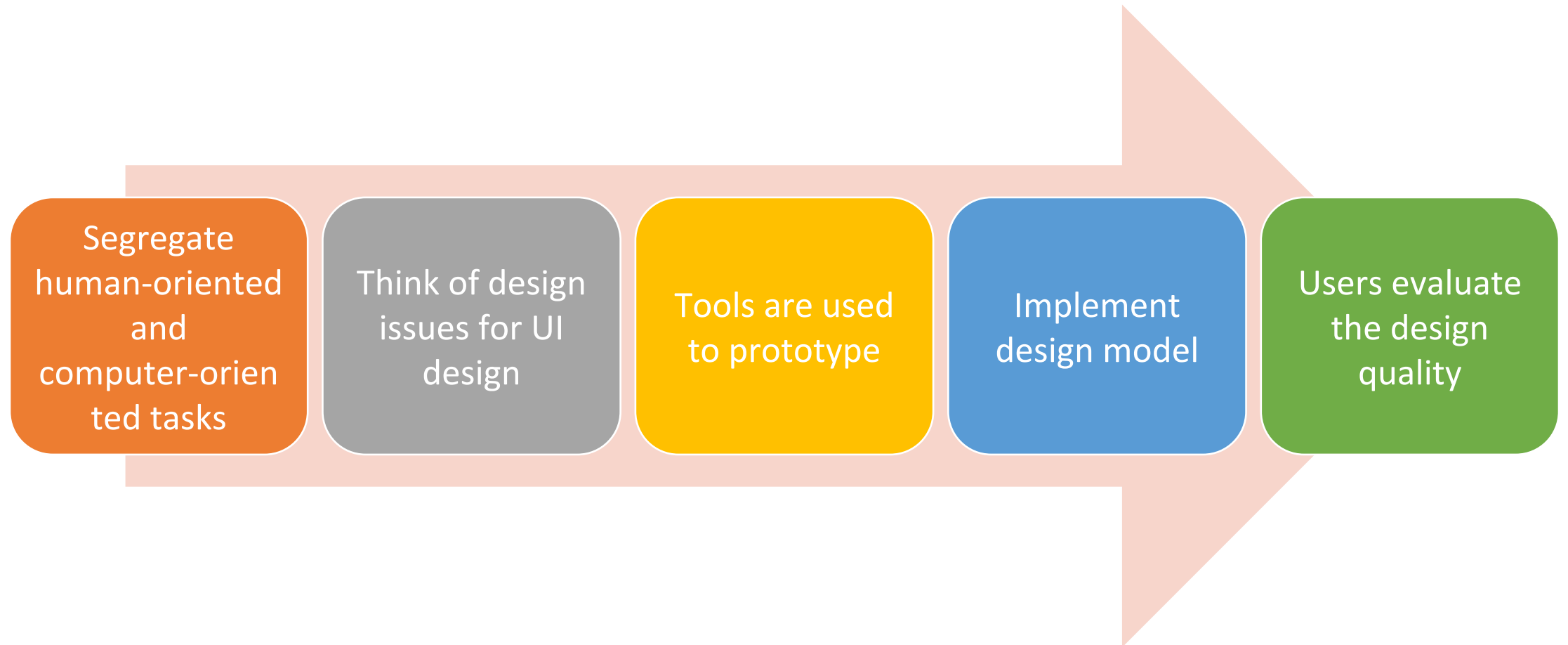
Principle 2: Reduce User's memory load

- **Define intuitive shortcuts**
 - Bind shortcuts that are easy to remember like ctrl+p for **print**
- **Visual layout should be near to real-time systems**
 - E.g., the bill paying process should be the same as in real life.
 - All steps must be fulfilled.
- **Disclose information in a progressive fashion**
 - Information displayed on multiple levels.
 - E.g., E-commerce websites have just pictures. Click that to show detailed information, different underline options shown when clicked on the list further

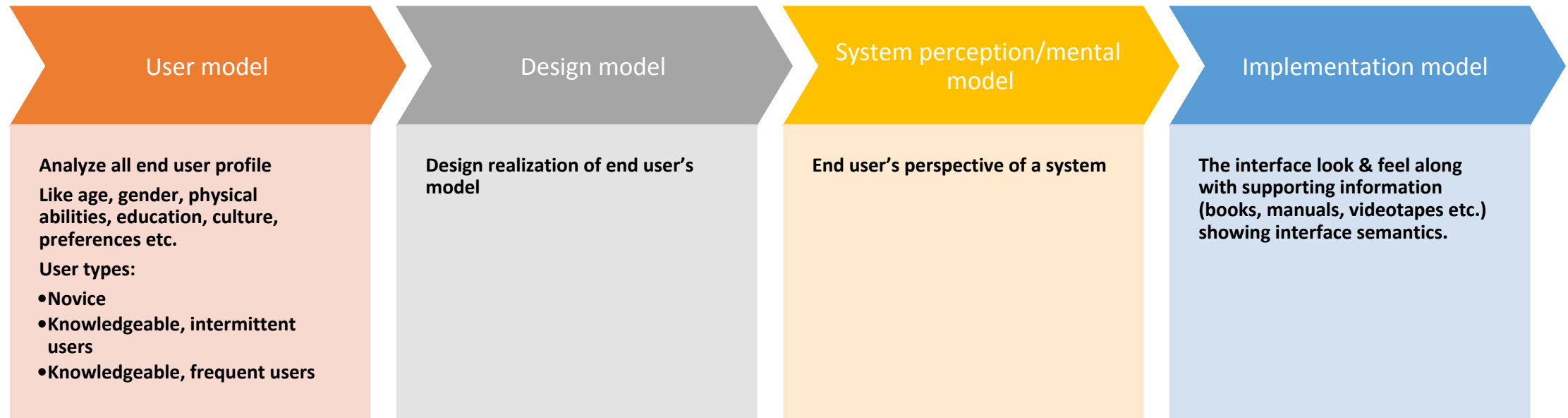
Principle 3: Make consistent interface

- **Allow users to put tasks in meaningful context**
 - User should have an idea of the work they are doing
 - **Like windows should be labeled with titles, consistent color coding should be used for the same tasks etc.**
- **Maintain consistency across a family of applications**
 - A set of applications must follow a particular pattern
 - As the front and the back view is same in whole MS office.
- **Don't make changes in system against user's previous habits, until you are compelled to do so**
 - Don't redefine the user's past experiences in UI
 - For example, Ctrl+S is used for save. Don't bind it with scaling. As user is expecting on his past experience that it'll save the file.

UI Analysis & design

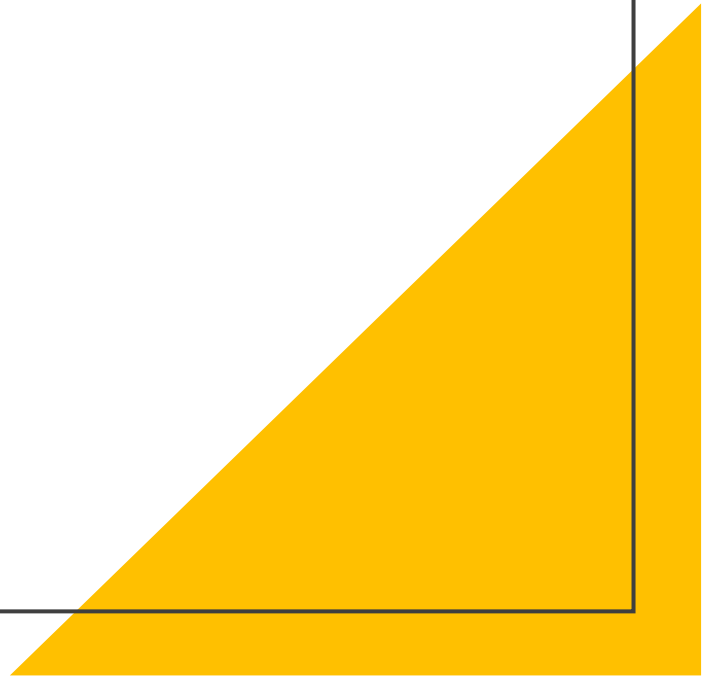


UI Design models



UI Design models

- Hence, an effective design is one in which the user mental model and implementation model coincides



UI analysis & design Process

- Iterative process ~ use spiral process model
- Four steps are:
 - Interface analysis & modeling
 - Interface design
 - Interface construction ~ involve prototyping approach
 - Interface validation
- Since its spiral model so these steps need to be processed again and again